

QUESTION 1

a) Description of the dataset

```
?crabs
```

The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b) Loading the data set and structure

```
crabs  
head(crabs)
```

	sp	sex	index	FL	RW	CL	CW	BD
1	B	M	1	8.1	6.7	16.1	19.0	7.0
2	B	M	2	8.8	7.7	18.1	20.8	7.4
3	B	M	3	9.2	7.8	19.0	22.4	7.7
4	B	M	4	9.6	7.9	20.1	23.1	8.2
5	B	M	5	9.8	8.0	20.3	23.0	8.2
6	B	M	6	10.8	9.0	23.0	26.5	9.8

Structure of the data

```
str(crabs)
```

```
data.frame': 200 obs. of 8 variables:  
 $ sp      : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...  
 $ sex     : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...  
 $ index   : int  1 2 3 4 5 6 7 8 9 10 ...  
 $ FL      : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...  
 $ RW      : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...  
 $ CL      : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2  
 ...  
 $ CW      : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3  
 ...  
 $ BD      : num  7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

The crabs data set is a data frame with 200 observations and 8 variables. it has 6 numeric variables and 2 categorical variables.

c)

```
f_table_sp.sex <- table(crabs$sp, crabs$sex)
print(f_table_sp.sex)
```

```
      F  M
B  50  50
O  50  50
```

Frequency table for species

```
f.table.sp<-table(crabs$sp)
print(f.table.sp)
```

```
B      O
100  100
```

Frequency for sex

```
f.table.sex<-table(crabs$sex)
print(f.table.sex)
```

```
F      M
100  100
```

d)Bar chart summarizing the group compositions

```
par(mar=c(4,4,3,1))
barplot.group<-barplot(f_table_sp.sex ,beside=TRUE,
xlab="Species", ylab="Frequency", main="Group Composition of
Crabs",col=3:4)
```

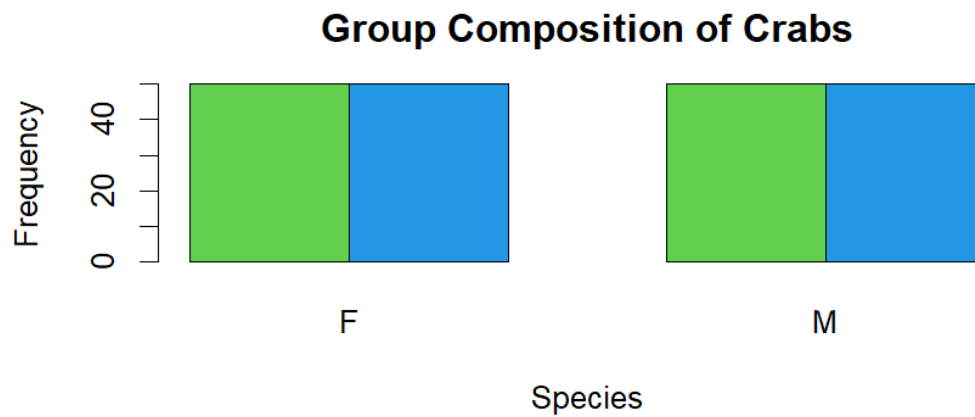


Figure 1.1. A bar chart on group compositions

e) The data set is balanced across all the four groups each combination has the same amount of crabs.

QUESTION 2

a)

```
##Boxplot of BD for four groups defined by species and sex

Boxplot_BD_sp_sex<-boxplot(crabs$BD[crabs$sp == "B" & crabs$sex
== "M"], crabs$BD[crabs$sp == "B" & crabs$sex ==
"F"],crabs$BD[crabs$sp == "O" & crabs$sex == "M"],
      crabs$BD[crabs$sp == "O" & crabs$sex == "F"],
      names = c("Blue M", "Blue F", "Orange M", "Orange F"),
      main = "Body Depth (BD) by Species and Sex",
      ylab = "Body Depth (mm)",col=2:6)
```

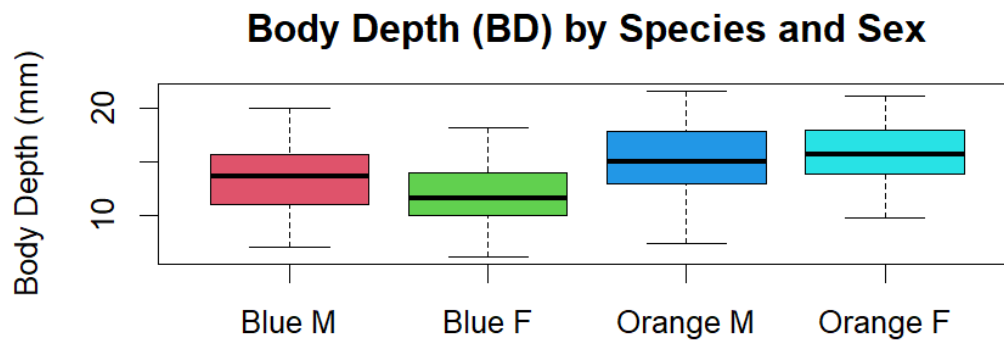


Figure 2. A box and whisker plot of body depth of the four groups by species and sex

There are no outliers in the data set. The data for all four compositions appear symmetric or normally distributed.

b)

```
###central tendency between groups by species and sex

list(crabs$species, crabs$sex)

##Mean
Mean_sp_sex<-tapply(crabs$BD, list(crabs$sp, crabs$sex), mean)
##Median
Median_sp_sex<-tapply(crabs$BD, list(crabs$sp, crabs$sex),
median)

##Variance
Variance_sp_sex<-tapply(crabs$BD, list(crabs$sp, crabs$sex),
var)
```

Mean

	F	M
B	11.816	13.350
O	15.632	15.324

Median

	F	M
B	11.60	13.6
O	15.65	15.0

Variability

	F	M
B	7.576065	10.23929
O	7.576914	12.44104

The mean and median for all the groups are approximately the same. Crabs that are orange and male have the largest variability in the data set and crabs that are blue and female an orange and female have the same variance.

QUESTION 3

a)

```
##Histogram of CL for each species
par(mfrow=c(1,2))
# Histogram for species B
Histogram_Blue<-hist(crabs$CL[crabs$sp == "B"],
  main = "",
  xlab = "Carapace Length (CL) for B",
  col = "lightblue")
# Histogram for species O
par(mfrow=c(1,2))
# Histogram for species B
Histogram_Orange<-hist(crabs$CL[crabs$sp == "O"],
  main = "",
  xlab = "Carapace Length (CL) for O",
  col = "red")
```

A Histogram of carapace length for both species

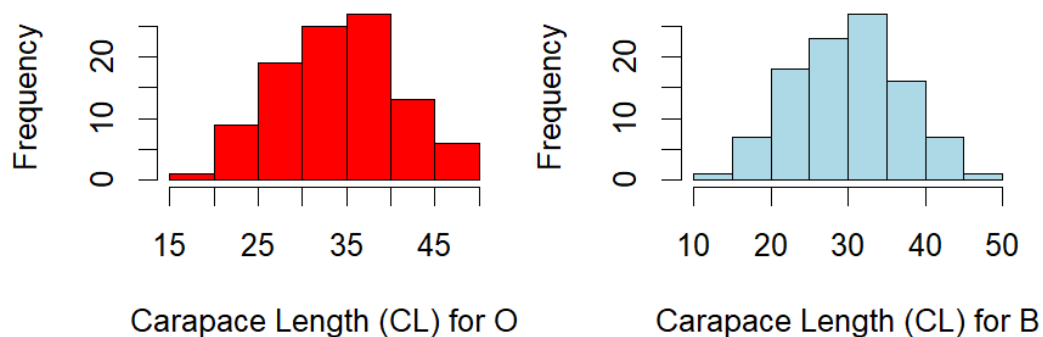


Figure 3 .A histogram of CL for both species

- c) Both datasets are unimodal, they all have a single modal value. The histogram for orange species appears to be symmetric to the left and the histogram for Blue appears to be slightly skewed to the right.

QUESTION 5

a)

```
###Scatter plot of the five measurements
par(mar=c(4,4,4,1))
pairs(crabs[, 4:8],
      main = "Scatter Plot of Crab Measurements",
      col = "black")
```

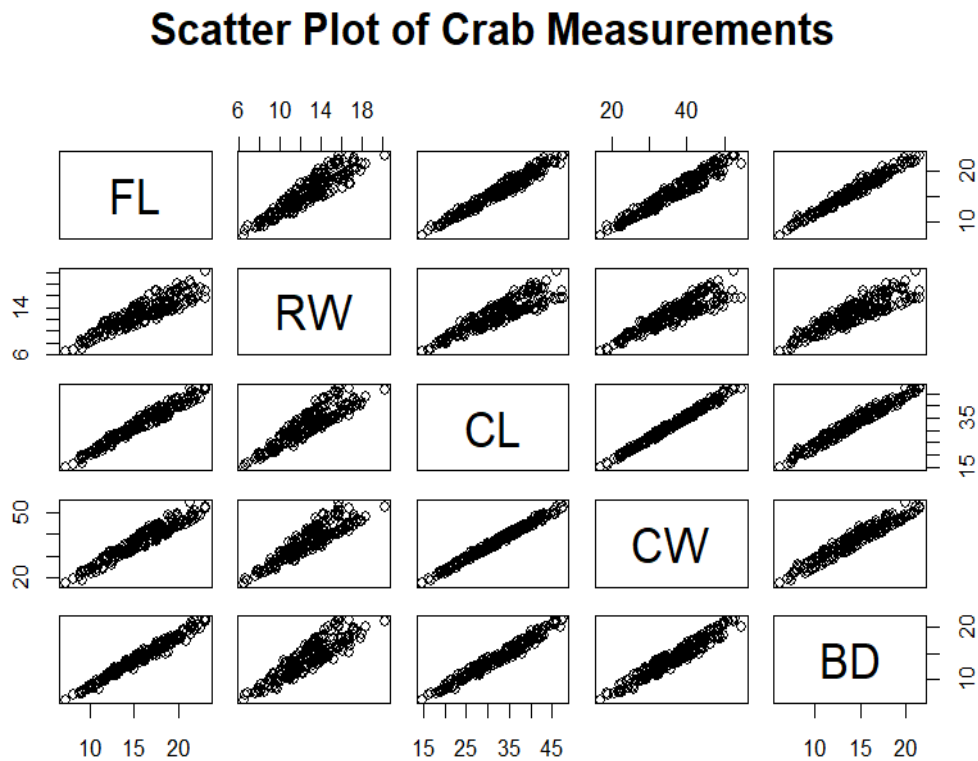


Figure 4. A scatter plot of the five measurements

b) There is a positive relationship between all the five measurements. The measurements are positively correlated.

QUESTION 4

a)

```
###Comparison of mean measurements

# Create 4 groups

B_F <- crabs[crabs$sp=="B" & crabs$sex=="F", ]
B_M <- crabs[crabs$sp=="B" & crabs$sex=="M", ]
O_F <- crabs[crabs$sp=="O" & crabs$sex=="F", ]
O_M <- crabs[crabs$sp=="O" & crabs$sex=="M", ]
means table <- data.frame(
  Group = c("B_F", "B_M", "O_F", "O_M"),
  FL = c(mean(B_F$FL), mean(B_M$FL), mean(O_F$FL),
mean(O_M$FL)),
  RW = c(mean(B_F$RW), mean(B_M$RW), mean(O_F$RW),
mean(O_M$RW)),
  CL = c(mean(B_F$CL), mean(B_M$CL), mean(O_F$CL),
mean(O_M$CL)),
  CW = c(mean(B_F$CW), mean(B_M$CW), mean(O_F$CW),
mean(O_M$CW)),
  BD = c(mean(B_F$BD), mean(B_M$BD), mean(O_F$BD),
mean(O_M$BD))
)
means table
```

Group	FL	RW	CL	CW	BD
1 B_F	13.270	12.138	28.102	32.624	11.816
2 B_M	14.842	11.718	32.014	36.810	13.350
3 O_F	17.594	14.836	34.618	39.036	15.632
4 O_M	16.626	12.262	33.688	37.188	15.324

b)

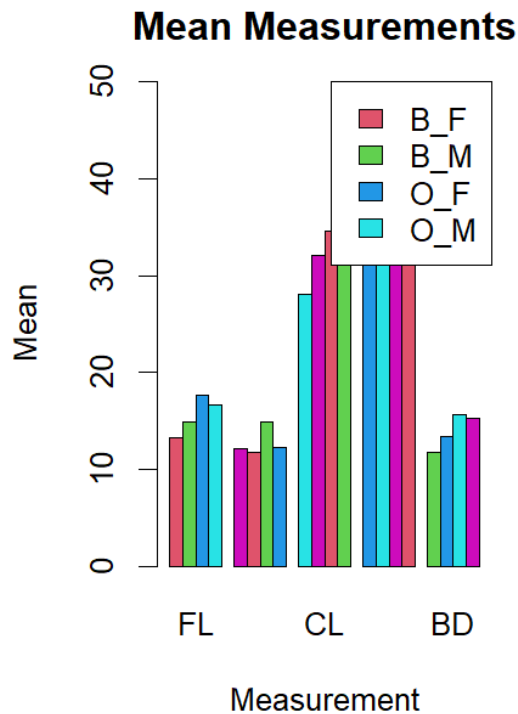
```
par(mar = c(4,4,3,1))
bar data <- as.matrix(means table[, 2:6])
rownames(bar data) <- means table$Group

barplot(bar data,
        beside = TRUE,
```

```

main = "Mean Measurements",
xlab = "Measurement",
ylab = "Mean",
legend.text = rownames(bar_data),
col = 2:6,
args.legend = list(x = "topright"), ylim = c(0, 50))

```



QUESTION 6

A COMPREHENSIVE REPORT ON THE DESCRIPTIVE ANALYSIS OF THE CRABS DATASET

The crabs data set contains 200 sampled from two species. It also records sex and five key measurements.

The bar chart in Figure 1 shows that the data is balanced and are of the same sample size. Across the four groups shown in the boxplot in Figure 2, males generally have a larger body depth compared to the females. The means and medians are also similar making the data symmetric. For Carapace length, the species B are slightly skewed to the right while species O appear to be symmetric, both data sets are unimodal, that is they have one distinct modal value.

The dataset shows clear **species and sex patterns**, with larger males and O species.